MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-561( )(V)/TTC-41(V)

This specification is approved for use by Communications & Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the Central Office, Telephone, Automatic AN/TTC-41(V) which is used to provide automatic central office switchboard facilities for Army Communication systems. (See 6.1 and 6.5)

1.2 Description.- Shelter, Electrical Equipment S-561( )(V)/TTC-41(V), hereinafter called the shelter facility, includes a modified S-250/G shelter and all racks, brackets, shelves, tie-downs, patch panel, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TTC-41(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards.- The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications & Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-Packaging Methods Of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronics and Associated Electrical Equipment)
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys. Nickle Alloys, Heat-Resisting Alloys & Titanium, Spot & Seam
MIL-T-7928	Terminals, Lug & Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-S-55507	Shelter, Electrical Equipment (with or without equipment), packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Prodecures & Tables for Inspection by attributes
MIL-STD-252	Wired Equipment, Classification of Visual & Mechanical Defects
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requiremnts for Electrical Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-1474	Noise Limits for Army material
MIL-STD-1857	Grounding, Bonding, and Shielding Design Practices

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2.1.2 Government drawings. The following Government drawings form a part of this specification ot the extend specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
Electronics Command

DL-SC-B-951705	Shelter, Electrical Equipment S-561( ) (V)1/TTC-41(V)
SC-D-913818	Power Schematic
SC-F-913819	Signal Schematic
DL-SC-B-951705	Shelter, Electric Equipment S-561( )(V)1/TTC-41(V)
SC-D-951733	Power Schematic S-561( )(V)1/TTC-41(V)
SC-D-883634	Power Schematic S-561( )(V)2/TTC-41(V)
SC-D-951734	Signal Schematic S-561( ) (V)1/TTC-41(V)
SC-D-88365	Signal Schematic S-561( )(V)2/TTC-41(V)

(Copies of specifications standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order or precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the test of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article.- When specified in the contract, the contractor shall furnish first article units in accordance with 4.3 .

3.2 Construction.- The shelter facility shall be constructed in accordance with the drawings as listed on DL-SC-B-864460 and the requirements specified herein. (See 3.25)

3.3 Electrical installation.- (See 4.6)

3.3.1 Power system wiring and components.- The complete in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- Continuity shall exist between each pin of the power entrance connectors and respective circuit terminations and shall be less than 3 ohms. (See 4.6.1 and 4.6.1.1)

3.3.1.2 Dielectric breakdown voltage.- The insulation of all power wires, cables and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than 30 seconds. (See 4.6.1 and 4.6.1.2)

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3.3.1.3 Insulation resistance.- The resistance of insulation between any two circuit conductors and also between each of the four ungrounded power line conductors and shelter facility ground terminal shall not be less than 100 megohms. (See 4.6.1 and 4.6.1.3)

3.3.2 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown in the power schematic drawing when tested in accordance with 4.6.2.

3.3.3 Signal system wiring and components.- The completed inplace signal system installation shall meet the following requirements:

3.3.3.1 Continuity.- Continuity shall exist between End terminations of all signal wires and shall be less than 5 ohms. (See 4.6.3 and 4.6.3.1)

3.3.3.2 Dielectric strength of insulation.- The insulation of all signal wires, cables, and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than 5 seconds. (See 4.6.3.2)

3.3.3.3 Insulation resistance.- The resistance of insulation of each signal line conductor and all other signal line conductors connected to ground shall not be less than 100 megohms. (See 4.6.3.3)

3.3.3.4 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pullout requirements of MIL-C-3885. (See 4.4)

3.3.3.5 Suppression of electromagnetic interference.- The motors in the ventilating equipment used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4)

3.4 Systems operation tests.- With all operational equipment of the AN/TTC-41(V) assemblage installed, the operational requirements of all power and signal cable assemblies, switches, connectors, and operational equipment shall be met when tested in accordance with 4.7.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicle without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.8)

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3.6 Railroad transportation.- The shelter facility shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail and shall be subjected to and withstand the tests of 4.9 without sustaining any permanent damage such as buckling, cracking, delamination or other permanent deformation.

3.7 Drop test.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.10)

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.11. During and after the test there shall be no leakage into the facility or its walls, floor, roof or doors.

3.9 Fording.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the facility or its walls, floor or doors.

3.10 Humidity.- Hardware sample shall have no evidence of errosion, rust or deterioration after testing. (See 4.13)

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage or permanent distortion to the shelter mounting surface or insert head or threads and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.12 Acoustic1 Noise.- The acoustical noise shall not exceed the criteria specified in MIL-STD-1474A(MI), Table 2, Category F. (See 4.15)

3.13 Cleaning.- (See 3.25)

3.13.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

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3.13.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware and other foreign material.

3.14 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 3.25 and 4.4)

3.15 Marking.- (See 3.25 and 4.4)

3.15.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.15.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.15.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The location of the centers-of-gravity shall be determined by the contractor using the first article of production.

3.15.4 Interior and exterior marking.- All interior and exterior markings shall be defined on drawings.

3.16 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of MIL-STD-454. (See 3.25 and 4.4)

3.17 Seal, water vapor.- All joints of the framing for the exhaust blower, power and signal entrance boxes, air conditioner and battery exhaust shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant Silastic, 738 as made by Dow Corning, or equal using the manufacturer recommended primer. (See 3.25 and 4.18)

3.18 Securing of parts.- Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.25 and 4.18)

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3.19 Soldering.- (See 3.25 and 4.18).

3.19.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.19.2 Acid or acid salts.- No acid or acid salts shall be used on preparation for or during soldering; however, exception permitted for preliminary tinning of electrical connections and for soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.19.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, bursh bristles, or other foreign materials remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.20 Tropicalization of material.- Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings as follows:

3.20.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.20.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.20.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to met specified requirements when subjected to specified service conditions.

3.20.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposed to use. (See 6.7)

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3.21 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-45205 CL.B as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag, inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.22 Wiring and cabling.- (See 3.25 and 4.18) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.22.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected or disconnected.

c. To facilitate field repair of broken or cut wires.

d. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.22.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.22.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.22.4 Solderless temrinal lug connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of MIL-T-7928. (See 4.4 and 4.16)

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3.22.5 Identification of wiring.- Wiring shall be identified by color coding and wire pair marking, as indicated on the drawings.

3.22.6 Cord Assemblies.- Power cable and cord assemblies shall be fabricated in accordance with the drawings and meet the requirements and tests of MIL-C-3885. (See 4.4)

3.22.7 Testing.- Power and signal wiring of the S-561()(V)/TTC-41(V) shall be subjected to and pass the tests described in 4.6.

3.23 Suppression of radio frequency interference.- The motors in the exhaust fans, air conditioner, the fluorescent lighting and emanation limits of MIL-STD-461 for tactical support equipment, if other than those items defined by the drawings. (See 4.4)

3.24 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.17) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.25 Workmanship.- (See 4.18) The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.2 Construction
3.3 Electrical installation
3.11 Mounting inserts
3.13 Cleaning
3.14 Finish
3.15 Marking
3.16 Riveting
3.17 Seal, water vapor
3.18 Securing of parts
3.19 Soldering
3.20 Tropicalization of materials
3.21 Welding
3.22 Wiring of cabling

3.26 Safety engineering.- The equipment design criteria shall be in accordance with Title 29, Code of Federal Regulations, Chapter XBII, part 1910, "Occupational Safety and Health Standards", Requirement 1, MIL-STD-454; and shall include but not be limited to the following:

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3.26.1 Personnel Hazards.- Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (See 4.18 and 4.21).

3.26.2 Edge Rounding Exposed.- Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

3.26.3 Environmental Safety.-

a. Personnel shall be provided an acoustical environment which will not cause personnel injury, or cause fatigue, or in any other way degrade overall system effectiveness.

b. No radioactive material shall be utilized without prior express approval of the Contracting Officer (potential sources of radiation include luminous dials/markings, electron tubes, surge arrestors and lenses).

c. A Carbon Monoxide (CO) detector/alarm shall be provided in a equipment shelter/assemblage when it will be occupied with the vehicle engine operation; or when a multifuel heater will be used. It shall effectively detect the threshold limit value (TLV) for CO as detailed in 09 CFR 1910.1000, Table Z-1. The CO detector must be reliable and shall provide an audio as well as visual alarm.

3.26.4 Exposed Voltage.- All contacts, terminals and like devices having voltages between 70 and 500 volts rms or dc with respect to ground, shall be guarded from accidental contact by personnel if such points are exposed to contact during direct support or operator maintenance.

3.26.5 Gounding.- The system design shall comply with the following provisions to assure all noncurrent-carrying conductors:

a. the continuity of a grounded conductor shall not be dependent upon device connections, such as lampholders, receptacles, etc., where the removal of such devices would interrupt the continuity.

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b. All power input or interconnecting cables shall provided with a green insulated equipment ground wire (EGW) used only for grounding noncurrent-carrying metal parts of equipment. Test equipment leads shall have provisions to insure that the chassis of equipment under test is also at ground potential.

c. The equipment ground wire shall be terminated at both ends in the same manner as the other conductors and shall be of such size and current rating as to conduct safely any fault currents that may be imposed upon it. It shall carry no current at any time other than during a ground fault.

d. The equipment ground wire shall not be connected to the ground power return (neutral) at any point other than the generator or the secondary of the power distribution transformer. (Example: SEPARATION OF NEUTRAL FROM SHELTER FRAME GROUND).

e. In all electrical and electronic equipment, the AC power neutral (white wire) shall be isolated from the equipment by a minimum of one (1) megohm DC resistance measured at the power input connector.

f. Color coding for all power cables and power distribution wiring to rack level shall comply with Para 3.2.8 of MIL-STD-1857(EL) and the following:

(1) Single phase a.c. service (such as 115 volt):

Black (hot)
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 3-wire cable required.

(2) Three phase a.c. circuit:

Black (hot) Phase A
Red (hot) Phase B
Blue (hot) Phase C
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 5-wire cable is required.

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(3) Single phase, tapped a.c. (such as 110/220):

Black (hot)
Red (hot)
White (neutral)
Green (equipment ground)

NOTE: Ground wire is essential; 4-wire cable required.

3.26.6 Chemical safety.- The system design shall include the following provisions:

a. No material or component which during any phase of the life cycle will evolve toxic gases, vapors, fumes, or droplets shall be used in the system which would detrimental to the performance of the equipment or the health of the equipment operator. The materials also shall not liberate gases which will produce an explosive atmosphere.

b. No treating materials shall be used that are identified in the Toxic Substance List (published by National Institute for Occupational Safety and Health) that will produce toxic effects via the respiratory tract, eye, skin (cutaneous and subcutaneous), or mouth. This includes handling the treated material during fabrication, transportation, operation or maintenance of the parts in which the treated materials are used or duing sue of the finished parts when used for the purpose intended.

4. QUALITY ASSURANCE PROVITIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance.- All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection.- The inspections shall be classified as follows:

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a. First article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection of equipment before packaging. (See 4.5)

d. Packaging inspection. (See 4.19)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units.- The contractor shall furnish one (1) first article unit of the complete S0561( )(V)/TTC-41(V).

4.3.2 First article inspection.- The first article inspeciton shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I - First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.24	4.17
3. Mounting Insert	3.11	4.14.1
4. Solderless-terminal-lug-connection	3.22.4	4.16
5. Humidity	3.10	4.13
6. Visual & Mechanical	3.25	4.18
7. Electrical Installation	3.3	4.6
8. System Operation	3.4	4.7
9. Acoustical Noise	3.12	4.15
10. Vehicular transportation	3.5	4.8
11. Railroad transportation	3.6	4.9
12. Drop tests	3.7	4.10
13. Watertightness	3.8	4.11
14. Fording	3.9	4.12
15. Repeat 7 - Electrical Installation	3.3	4.6
16. Repeat 8 - System operation	3.4	4.7

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where required
Cable & Cord Assemblies	3.3.3.4, 3.22.6
Finish	3.14
Interference Suppression	3.3.3.5
Marking	3.15
Riveting	3.16
Solderless terminal lug connections	3.22.4
Welding	3.21

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4.5 Quality conformance inspection of equipment before packaging.- The contractor shall perform the inspection specified in paragraph 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in order listed in Table II.

TABLE II Group A Inspection

Inspection	Reqt Para	Test Para	AQL%	
			Major	Minor
Visual & Mechanical	3.25	4.18	1.0	4.0
Electrical Installation	3.3	4.6	1.0	*

* All defects are major.

4.5.2 Group B Inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans.- The Group B sampling plans for the AQL's listed in Table III shall be as follows:

AQL	Inspection Level
4.0%	S-4
2.5%	S-4

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4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in the order listed in Table III.

TABLE III - Group B Inspection

Inspection	Reqt Para	Insp Para	AQL% (See 4.5.2.1)
Interchangeability	3.24	4.17	4.0
Mounting Insert	3.11	4.14.2	4.0
Watertightness	3.8	4.11	2.5
Fording	3.9	4.12	2.5

4.5.3 Group C Inspection.- This inspection shall be as listed in Table IV.

TABLE IV - Group C Inspection

Inspection	Reqt Para	Insp Para
Solderless terminal lug connection	3.22.4	4.16
Mounting inserts	3.11	4.14.1
System operation	3.4	4.7
Acoustical noise	3.12	4.15
Vehicular transportation	3.5	4.8
Railroad transportation	3.6	4.9
Drop tests	3.7	4.10
Humidity	3.10	4.13

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table IV. Units subjected to Group C Drop Test shall subsequently be subject to and meet the watertightness and Fording Test of 4.11 and 4.12.

4.5.3.1 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

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4.5.3.2 Sampling for inspection of Drop Tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the truck and rail transportation tests of 4.5.3.3. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.3 Sampling for other inspection.- For inspection tests listed below one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the Government.

a. Mounting inserts

b. System operation

c. Acoustical noise

d. Vehicular transportation

e. Railroad transportation

f. Humidity

4.5.3.4 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection.

4.5.5 Alternate Group A Audit and Group B inspection procedures.- The contractor may elect to use a continous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Sampling plans for the AQL's listed in Table II Group A inspection and Table III Group B inspection shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production contractor shall notify the contracting officer in writing of which method he intends to use. The method may not be changed during production without written approval of the Government.

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4.6 Electrical Installation Tests.- The sehlter facility shall be subjected to the tests of 4.6.1 through 4.6.3 below. Before applying power to the shelter facility, an electrical ground connection at the power entrance box shall be made. Power to be supplied shall be 115 VAC, 60 Hertz.

4.6.1 Power systems and components installation tests.- Refer to the applicable Power Schematic Plate Drawing.

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent and neon lamps.

(2) All fluorescent lamp starters.

(3) Blowers, heaters, air conditioners and battery exhaust fan.

b. Remove power distribution box cover panel and disconnect conductors from the voltmeter terminals. Position or isolate the disconnected leads so as not to short to each other or facility ground.

c. Remove DC Lights Relay and place a jumper across pins 1 and 3 and across pins 6 and 8.

4.6.1.1 Continuity of conductors.- There shall be point-to-point circuit continuity on all power line circuits and the resistance of the continuous circuit shall not exceed 3 ohms. Continuity resistance tests which meet this requirement shall be performed on each power line circuit in accordance with the following paragraphs. A calibrated low resistance ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position:

(1) All circuit breakers.

(2) Fluorescent lamp switch.

(3) Incandescent lamp switch.

(4) Blackout bypass switch (to bypass position).

(5) Blower switches.

b. Measure continuity resistance from pin 1 of 115V power connector IN located at the power entrance box to each of the following:

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(1) Brass (hot) contacts of all convenience, intercom, heater, air conditioner, power supply and blower power receptacles.

(2) Black wire termination at each fluorescent and incandescent lamp. Continuity shall exist, the Resistance Requirement is waived.

(3) Pin 2 of DC Lights Relay.

(4) Pin 1 of 115V power connector OUT.

(5) Pin 2 and 4 of 115V Power Connector OUT, no continuity shall exist.

c. Place all controls listed in step a in the OFF position, except main circuit breaker.

d. Repeat step b measuring continuity resistance from pin 2 of 115V power connector IN to the silver contacts of each power receptacle white wire termination at each fluorescent and incandescent lamp, pin for DC Lights Relays and pin 2 of 115V power connector OUT.

e. Place main circuit breaker in the OFF position.

f. Measure continuity resistance from pin 4 (ground) of 115V power connector IN to each of the following:

(1) Ground Terminal at each power receptacle.

(2) Ground Terminal located at power entrance box, power distribution box, equipment racks and raceways.

(3) Pin 4 of 115V power connector OUT.

(4) Pin 2 of 115V power connector IN, no continuity shall exist.

NOTE: The following procedures apply to (V2) shelters, for V1 shelters the procedures are the same except there is no DC light switch, no switchboards 4 and 5 and no PP-6224 No. 2.

g. Place the DC lights switch in the Battery 1 position.

h. Measure continuity resistance from Pin 4 of the DC lights relay to each of the following:

(1) Pin A of Switchboards 1, 2 and 3 DC Power cable.

(2) Pin A of PP-6224 No. 1 DC Power Cable.

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(3) Pin A of PP-6224 No. 2 DC Power Cable, no continuity shall exist.

(4) Pin B of both PP-6224 DC Power Cables, no continuity shall exist.

i. Place the DC lights switch in the Battery 2 position.

j. Measure continuity resistance form Pin 4 of the DC lights relay to each of the following:

(1) Pin A of Switchboard 4 and 5 DC Power Cable.

(2) Pin A of PP-6224 No. 1 DC Power Cable.

(3) Pin B of both PP-6224 DC Power Cable.

k. Measure continuity resistance between Pin 5 of the DC lights relay and the red wire termination at each DC lamp fixture.

l. Measure continuity resistance from the black wire terminations of the DC lamp fixtures to each of the following:

(1) Pin B of Switchboard 4 and 5 DC Power Cable.

(2) Pin B of PP-6224 No. 2 DC Power Cable.

m. Place the DC lights switch in the Battery 1 position.

n. Measure continuity resistance between the black wire termination of the DC lamp fixtures to each of the following:

(1) Pin B of Switchboards 1, 2 and 3 DC Power Cable.

(2) Pin B of PP-6224 No. 1 DC Power Cable.

4.6.1.2. Dielectric breakdown voltage.- The dielectric strength of insulation of all power wires, cables,and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds. A non-destructive DC dielectric tester shall be used in performing the tests. Test procedures shall be as follows:

CAUTION: Take preventative measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test in order to eliminate the danger of electrical shock.

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a. Place all circuit controls listed under 4.6.1.1.a in the ON position except the fluorescent lamp switch.

b. Test between pin 1 and pin 4 of 115V power connector IN. Adjust tester to an output value of 500 volts DC for not less than 30 seconds. The dielectric strength requirements shall be met.

c. Place fluorescent lamp switch in the ON position.

d. Repeat step b with the tester connected first across pins 1 and 2 and then across pins 2 and 4 of 115V power connector IN.

4.6.1.3 Insulation resistance.- The insulation resistance between any two of three circuit conductors and also between each of the two ungrounded power line conductors and shelter facility ground shall not be less than 100 megohms when tested in accordance with the following procedures. A megohmeter or equivalent shall be used in performing the tests.

a. Place all circuit controls listed under 4.6.1.1.a in the ON position except the blackout bypass switch and fluorescent lamp switch.

b. Close shelter door securely.

c. Test between pins 1 and 4 of 115V power connector IN. The tester output voltage shall be between 100 and 500 volts DC. Maintain this potential until the meter indicator has ceased fulctuating and indicates a steady or increasing insulation resistance. The insulation resistance requirement shall be met.

d. Place the blackout bypass switch in the ON position.

e. Repeat step c with the tester connected first across pins 1 and 2 and then across pins 2 and 4 of 115V power connector IN.

4.6.2 Power Distributional and Operational Function Tests.- Power distribution and operational function tests shall be peformed on each power line circuit in accordance with the following paragraphs.

4.6.2.1 Test Preparation.- (Refer to power schematic plate)

a. Replace or reconnect voltmeter and all electrical loads which were removed or disconnected by 4.6.1.a, 4.6.1.b and 4.6.1.c.

b. Close power distribution box panel cover.

c. Place all circuit breakers and switches in the OFF position.

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d. Place a 115 volt AC test lamp with grounding plug into all convenience, heater, power supply, air conditioner and intercom receptacles.

e. Open outer exhaust port cover of blower and insert AC line plug of blower into appropriate receptacle.

f. Open battery exhaust vent cover.

4.6.2.2 Test for 115 volt AC Prime Power.-

a. Apply 115 volt AC 60 hertz, single phase prime power to the shelter facility at 115V power connector IN. The power indicator lamp shall light.

b. Measure line voltage by connecting a calibrated AC meter (0 - 150 volt scale) between pins 1 and 2 of 115 VAC power connector OUT.

c. Read voltage indicated on shelter facility voltmeter. The voltage read shall be within $\pm$ 2% of the voltage measured in step b above.

d. Place main circuit breaker in the ON position.

4.6.2.3 Tests of Circuit Breakers and Switches.-

Power Supply (supplies) Circuit Breaker (CB 1).-

a. Place circuit breaker CB 1 in the ON position. Neon lamps above CB 1 and test lamps in power supply receptacle shall light and battery exhaust fan shall operate.

b. Close battery exhaust vent cover. Battery exhaust fan audible alarm shall operate.

c. Open battery exhaust vent cover. Battery exhaust fan audible alarm shall cease operating.

d. Place CB 1 in the OFF position. All lamps mentioned in step a above shall go out and battery exhaust fan shall cease to operate.

Lights Circuit Breaker (CB 2).-

a. Place circuit breaker CB 2 in the ON position. Neon lamps above CB 2 shall light.

b. Place blackout bypass switch and fluorescent lamp switch in the ON position. All fluorescent lamps shall light.

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c. Place incandescent lamp switch in the ON position. All incandescent lamps shall light.

d. Place CB 2 in the OFF position. All lamps mentioned in steps a thru c shall go out.

e. Replace CB 2 in the ON position to restore light.

f. With shelter door open, place blackout bypass switch in the OFF position. All fluorescent and incandescent lamps shall go out.

g. Close shelter door. All fluorescent and incandescent lamps shall light.

h. Place fluorescent lamp switch in the OFF position. All fluorescent lamps shall go out.

i. Place incandescent lamp switch in the OFF position. All incandescent lamps shall go out.

j. Open shelter door and restore lights by placing blackout bypass switch, fluorescent light switch and incadescent lamp switch in the ON position.

Heater No. 1 Circuit Breaker (CB 5A).-

a. Place circuit breaker CB 5A (upper breaker of tandem circuit breaker CB 5A & 5B in the ON position. The upper neon lamp above tandem circuit breaker CB 5A & 5B and test lamp in Heater No. 1 receptacle shall light.

b. Place CB 5A in the OFF position. The lamps mentioned in step a shall go out.

Intercom and Convenience Circuit Breaker (CB 6A).-

a. Place circuit breaker CB 6A (upper breaker of tandem circuit breaker CB 6A & 6B in the ON position. The upper neon lamp above tandem circuit breaker CB 6A & 6B and test lamps in Intercom and Convenience 1 & 2 receptacles shall light.

b. Place 6A in the OFF position. The lamps mentioned in step a shall go out.

Heater No. 2 Circuit Breaker (CB 5B).- Repeat steps a and b of Heater No. 1 circuit breaker (CB 5A) test for CB 5 (lower breaker of tandem breakers CB 5A & 5B).

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Spare Circuit Breaker (CB 6B).-

a. Place circuit breaker CB 6B (lower breaker of tandem circuit breaker CB 6A & 6B in the ON position. The lower neon lamp above tandem circuit breaker CB 6A & 6B shall light.

b. Place CB 6B in the OFF position. The neon lamp above CB 6B shall go out.

Blower Circuit Breaker (CB 4).-

a. Place circuit breaker CB 4 in the ON position. The neon lamp above CB 4 shall light.

b. Place blower switch in the ON position. Blower shall operate and shall exhaust air from the shelter.

c. Place CB 4 in the OFF position. The neon lamp above CB 4 shall go out, the blower shall cease to operate.

Air Conditioner Circuit Breaker (CB 3).-

a. Place CB 3 in the ON position. The neon lamp above CB 3 and test lamp in air conditioner receptacle shall light.

b. Place CB 3 in the OFF position. The lamps mentioned in step a above shall go out.

Main Circuit Breaker (CB 7).-

a. Place all circuit breakers and switches in the ON position, all lamps except DC lamps shall light. The blower and battery exhaust fan shall operate.

b. Place main circuit breaker in the OFF postion. All lamps shall go out and the blower and battery exhaust fan shall cease to operate.

c. Place all circuit breakers in the OFF position except the lights circuit breaker, light switches and blackout bypass switch.

d. Place the main circuit breaker in the ON position. All fluorescent andincandescent lamps shall light.

Outside Convenience Circuit Breaker (CB 8).-

a. Place circuit breaker CB 8 in the ON position. Test Lamp in outside convenience receptacle shall light.

b. Place CB 8 in the OFF position. Test lamp shall go out.

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4.6.3 Signal Wiring and Components Installation Tests.- Refer to the Signal Wiring Diagram SC-D-883635.

4.6.3.1 Continuity of conductors.- There shall be circuit continuity between both end terminatinos of all signal line conductors and the resistance of the continuous circuit shall not exceed 5 ohms when tested in accordance with the following procedures. A calibrated low resistance reading ohmmeter or equivalent shall be used.

a. Connect the tester between contact 1a of U-187 connector for Signal 1 and contact w of corresponding MS-27473 connector. Circuit continuity shall exist and shall meet the continuity resistance requirement.

b. Repeat step a for contact 1B thru 24B of U-187 connector for SIG 1. In each case, connect the tester between corresponding contact of U-187 and M27473 connectors. See Table V.

c. Insert short circuit plug into CH 1 jack of SIG 1 on the switchboard side patch panel.

d. Connect tester between contact 1A of U-187 connector for SIG 1 and contact w of corresponding MS-27473 connector. No continuity shall exist.

e. Connect tester between contact 1B of U-187 connector for SIG 1 and contact N of corresponding MS-27473 connector. No continuity shall exist.

f. Connect tester between contacts w and N of appropriate MS-27473 connector for Switchboard 1. Signal continuity shall exist and meet the continuity requirement.

g. Connect tester between contacts 1A and 1B of U-187 connector for SIG 1. The tester shall indicate a resistance of 620 ohms.

TABLE V Continuity Checks

U-187	MS-27473 UPPER CONNECTOR	MS-27473 LOWER CONNECTOR
SIG 1, 3, 5, 7, 9		
1a	w	
1b	N	
2a	e	

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TABLE V Continuity Checks (continued)

U-187	MS-27473 UPPER CONNECTOR	MS-27473 LOWER CONNECTOR
SIG 1, 3, 5, 7, 9 (Cont'd):		
2b	h	
3a	J	
3b	L	
4a	H	
4b	u	
5a	b	
5b	s	
6a		h
6b		N
7a		L
7b		w
8a		K
8b		H
9a		e
9b		b
10a		u
10b		s
11a	i	
11b	x	
12a	M	
12b	FF	
13a	K	
13b	f	
14a	v	
14b	g	
15a	G	
15b	c	
16a	d	
16b	W	
17a	F	
17b	t	
18a	DD	
18b	Z	
19a		i
19b		x
20a		FF
20b		m
21a		M
21b		g

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TABLE V Continuity Checks (continued)

U-187	MS-27473 UPPER CONNECTOR	MS-27473 LOWER CONNECTOR
SIG 1, 3, 5, 7, 9 (Cont'd)		
22a		s
22b		AA
23a		J
23b		S
24a		HH
24b		BB
1a		G
1b		DD
2a		v
2b		q
3a		t
3b		f
4a		d
4b		X
5a	T	
5b	k	
6a	S	
6b	GG	
7a	m	
7b	z	
8a	j	
8b	L	
9a	p	
9b	U	
10a	V	
10b	n	
11a	CC	
11b	X	
12a	B	
12b	A	
13a	r	
13b	Y	
14a	D	
14b	C	
15a	E	
15b		T
16a		k
16b		S
		j

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TABLE V Continuity Checks (continued)

U-187	MS-27473 UPPER CONNECTOR	MS-27473 LOWER CONNECTOR
SIG 1, 3, 5, 7, 9 (Cont'd)		
17a		U
17b		z
18a		GG
18b		n
19a		W
19b		CC
20a		r
20b		V
21a		Z
21b		B
22a		E
22b		p
23a		Y
23b		D
24a		C
24b		A

h. Repeat steps c thru g for all other SIG 1 channel jacks on switchboard side patch panel. Refer to Table VI for sample tester connections and signal schematic for all other tester connections.

TABLE VI Sample Tester Connections

Short Circuited Jack	Tester U-187	Connection Pins MS-27473	Test Indication
CH. 1 2W	1A	w	No continuity
CH. 1 2W	1B	N	No continuity
CH. 1 2W		w,N	Continuity
CH. 1 2W	1A,1B		620 Ohms
CH. 2 2W	2A	e	No continuity
CH. 2 2W	2B	h	No continuity
CH. 2 2W		e,h	Continuity

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TABLE VI Sample Tester Connections (continued)

Short Circuited Jack	Tester U-187	Connection Pins MS-27473	Test Indication
CH. 13 2W/4WR	13A	K	No continuity
CH. 13 2W/4WR	13B	f	No continuity
CH. 13 2W/4WR		B,c,K,f	Continuity
CH. 13 2W/4WR	13A,13B		620 Ohms
CH. 13 4WT	14A	v	No continuity
CH. 13 4WT	14B	q	No continuity
CH. 13 4WT		v,q	Continuity
CH. 13 4WT	14A,14B		620 Ohms

i. Insert short circuit plug into CH. 1 jack of SIG 1 on line side patch panel.

j. Repeat steps d and e.

k. Connect tester between contacts w and N of MS-27473 connector for SIG 1. The tester shall indicate a resistance of 620 ohms.

l. Connect tester between contacts 1A and 1B of U-187 connector for SIG 1. Circuit continuity shall exist and meet the continuity requirement.

m. Repeat steps i thru l for all other SIG 1 channel jacks on line side patch panel.

n. Repeat steps a thru m for all other SIGNALS. See Table V for test connections.

o. Insert short circuit plug into SIG 1 phone jack at patch panel.

p. Connect tester between the following contacts. In each case, circuit continuity shall exist and meet the continuity requirements.

(1) Contacts 25A and 25B of U-187 connector for SIG 1.

(2) Contact 26A of U-187 connector for SIG 1 and intercom binding post A at power entrance box.

(3) Contact 26B of U-187 connector for SIG 1 and intercom ginding post B.

(4) Contact 26A of U-187 connector for SIG 1 and contact 26A of every other U-187 connector.

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(5) Contact 26B of U-187 connector for SIG 1 and contact 26B of every other U-187 connecotr.

q. Repeat steps o thru p (3) for all other signals.

r. Insert short circuit plug into TA-312 phone jack at patch panel.

s. Connect the tester between TA-312 phone binding post A and B at power entrance box. Circuit continuity shall exist and meet the continuity requirement.

t. Repeat steps r and s for TA-938 phone.

u. Insert a short circuit plug into the OPER Jack and SWBD 1 at the patch panel. Connect tester between pins g and AA of upper MS-27473 connector for SWBD 1 continuity shall exist and shall meet the continuity requirement.

v. Repeat step u for the TEST jack and pins HH and BB of the MS-27473 connector.

w. Repeat steps u and v for all other OPER and TEST jacks.

4.6.3.2 Dielectric Breakdown Tests.- The dielectric strength of insulation of all signal wires, cables and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds when tested in accordance with the following procedures. A non-dielectric test instrument and suitable test jigs shall be used (an automatic tester may be used).

a. By using appropriate test jigs and test fixtures, connect the following to shelter facility ground terminal:

(1) All contacts, 1A thru 26B of all signal connectors.

(2) All binding posts at the power entrance box.

b. Remove ground connection from contact 1A of SIG 1 connector. Connect tester between contact 1A and shelter facility ground.

c. Adjust tester to 500 volts DC output and maintain this potential for at least 5 seconds. The dielectric strength requirements shall be met. Re-establish ground connection to contact under test at completion of test.

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d. Repeat steps b and c for all other contacts of SIG 1 connector.

e. Repeat steps b thru d for all other SIGNALS.

f. Remove ground connection from TA-312 phone binding post A at power entrance box. Connect tester between binding post A and shelter facility ground.

g. Repeat step C, with a output potential of 250 volts DC.

h. Repeat steps f and g for all other phone and intercom binding posts. In each case, re-establish ground connection to contact under test at completion of test.

4.6.3.3 Insulation resistance.- The resistance of insulation of each singal line conductor and all other signal line conductors connected to ground shall not be less than 100 megohms when tested in accordance with the following procedures. A megohmmeter or equivalent shall be used in performing the tests. (An automatic tester may be used).

a. Repeat procedures 4.7.2.2 a thru h except c which shall be repeated by step b below.

b. Adjust tester to an output potential between 100 and 500 volts DC. Maintain this potential until the indicating meter has ceased to fluctuate and indicates a steady or increasing insulation resistance. The insulation resistance requirement shall be met. Re-establish ground connection to contact under test.

4.7 Systems Tests.- Note that the following Test Preparations are for Switchboard No. 1 thru 5. These preparations may be modified in accordance with the number of switchboards actually installed.

Test Preparaton.-

a. Install all operational equipments in their respective rack locations.

b. Connect all signal cables to designated connectors according to signal wiring diagram.

c. Connect power supply DC cable(s) from Terminal Box No. 1 to the respective power supply (supplies).

d. Connect battery cable(s) from Terminal Box No. 1 to the respective battery (batteries).

e. Connect power cables from Terminal Box No. 2 to the respective switchboards.

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f. Connect interconnection cables among switchboards as required.

g. Place MASTER SLAVE switch of switchboard No. 1 (switchboard No. 1 and No. 4) in the MASTER position. Place MASTER SLAVE switch of all other switchboards in the SLAVE position.

h. Place No. PLAN switch of switchboard No. 1 (switchboard No. 1 and No. 4) in the 1-30 position. Place No. PLAN switch of switchboard No. 2 (switchboards No. 2 and No. 5) in the 31-60 position. Place NO PLAN switch of switchboard NO. 3 in the 61-80 position.

i. Place TRUNK HUNT CONT. switches of all switchboards in the OFF position.

j. Place CALL PROCESS-PROGRAM switch of all switchboards in the CALL PROCESS position.

k. Connect beadset(s) to the master switchboard(s).

l. Connect all electrical equipment to their respective power receptacles.

m. Place power supply circuit breaker CB 1 in the ON position.

n. Place the AC and DC power switches of the power supply (supplies) in the ON position. Power indicator lamps shall light.

o. Place the battery switch (switches) in the ON position.

p. Place the power switch of all switchboards in the ON position. Power indicator lamps shall light. Battery ammeter(s) shall indicate current drain of approximately 1 ampere per switchboard.

q. Place DC light switch in Battery 1 position.

4.7.1 DC Power Circuit Tests.-

a. Place main circuit breaker in the OFF position. All AC lamps shall go out. All DC (emergency) lamps shall light. AC power indicator lamps on the power supply (supplies) shall go out. The battery (batteries) shall supply power to the DC lamps and all switchboards whose indicator lamps shall remain on. Battery ammeter(s) shall indicate a current drain of approximately 1 ampere per switchboard.

b. Place switch for battery No. 1 and Power Supply No. 1 AC and DC power switches in the OFF position. Power indicator lamps on switchboards No. 1, No. 2 and No. 3 shall go out. DC lamps shall also go out.

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c. Place DC light switch in the Battery 2 position. For 120 and 150 line assemblages DC lamps shall light. In 30, 60 and 90 line assemblages DC lamps shall remain out.

d. Place switch for battery No. 2 and Power Supply No. 2 AC and DC power switches in the OFF positions. Power indicator lamps on Switchboards No. 4 and No. 5 shall go out. The DC lamps shall also go out.

e. Place power switches on all switchboards in the OFF position.

f. Place battery switch (switches) in the ON position DC lamps shall light.

g. Place power switches on all switchboards in the ON position. Power indicator lamps on all switchboards shall light.

h. Place DC light switch in the OFF position. DC lamps shall go out.

i. Place DC light switch in the Battery 1 position. DC lamps shall light.

j. Place main circuit breaker in the ON position. Place power supply AC and DC switches in the ON position. AC lamps shall light, DC lamps shall go out.

k. Place battery switch (switches) in the OFF position. Power indicator lamps on all switchboards shall remain on.

l. Place battery switch (switches) in the ON position.

4.7.2 Switchboard Operational Tests.- The test procedures are based on the following signal Printed Circuit Board (PCB) card assignments in each switchboard:

a. Type 1 cards in Ch 1 thru Ch 5 (Ch 31 thru Ch 36) (Ch 61 thru Ch 66) card slots.

b. Type 2 cards in Ch 7 thru Ch 20 (Ch 37 thru Ch 50) (Ch 67 thru Ch 80) card slots.

c. Type 3 cards in Ch 21 thru Ch 30 (Ch 51 thru Ch 60) (Ch 81 thru Ch 90) card slots.

If the above PCB card assignments are not used, the following procedures may be modified as required. Para 4.7.2.1 pertains to TYPE I cards. Para 4.7.2.2 and 4.7.2.3 pertain to TYPE II cards and Para 4.7.2.4 pertains to TYPE III cards.

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4.7.2.1 Manual Switching of Two Wire Telephone Lines.-

a. Connect J-1077 junction boxes to each U-187 signal connector at signal entrance box.

b. Connect TA-312 telephones to CH 1 and CH 2 pins of Sig 1 junction box, i.e. to pins 1A, 1B and pins 2A and 2B respectively. Use Ch 1 telephone to call Ch 2 telephone by cranking it. The CALL-ANS button on the MASTER switchboard shall light. The operator completes a voice path with the caller by depressing the CALL-ANS button. The operator keys in IXX (102) on the switchboard keysender, where I is any number from 1 to 6, and XX is the called channel number. Voice patch between the Ch 1 and Ch 2 telephones shall be completed, and communications between them shall be possible. The operator exits by depressing either the OPR RLSE or CAN ANS button.

c. Repeat steps (b) with the roles of the telephones interchanged.

d. Repeat steps b and c for Ch 3 & Ch 4 and Ch 5 & 6 of Sig 1.

e. Repeat steps a thru d for all other odd number SIGNALS connected to a switchboard (Note the channel numbers for each switchboard).

4.7.2.2 Automatic Switching of Two-Wire Telephone Lines.-

a. Connect TA-938 telephones to Ch 7 and Ch 8 pins of SIG 1 junction box, i.e. to pins 7A, 7B and pins 8A and 8B respectively. Use Ch 7 telephone to call Ch 8 telephone by keying (dialing) IXX. The called telephone shall ring. Pick up telephone, a voice patch shall be completed, and communication between the telephones shall be possible.

b. Repeat step (a) with the roles of the telephone interchanged.

c. Repeat steps a and be for Ch 8 & Ch 10, 11 & 12 of SIG 1.

d. Connect TA-938 telephones to each pair of pins in SIG 1 and SIG 2 junction boxes as shown in Table VI. Patch between the corresponding pair of patch panel jacks. Use one telephone to call the other and vice versa. In each case, key in (dial) IXX to initiate call. The called telephone shall ring. Pick up telephone, a voice patch shall be completed and communication between the telephone shall be possible.

e. Repeat steps a thru d for all other SIGNALS connected to a switchboard.

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4.7.2.3 Switchboard Interconnections.-

a. Connect TA-938 telephones respectively to CH 9 and CH 39 of SIG 1 and SIG 3 junction boxes, i.e. to pins 9A and 9B in each junction box. Use the SIG telephone to call the SIG 3 telephone by keying (dialing) IXX (139). The called telephone shall ring. Pick up telephone. A voice patch shall be completed and communication between the telephones shall be possible.

TABLE VII

TELEPHONE CONNECTIONS			PATCH PANEL CONNECTIONS	
SIG	PHONE X PINS	PHONE Y PINS	LINE SIDE JACK	SWITCHBOARD SIDE JACK
1	13A, 13B	14A, 14B	CH 13, 4WT	CH 14, 2W/4WT
1	15A, 15B	16A, 16B	CH 14, 2W/4WR CH 14, 4WT	CH 15, 2W/4WT
1	17A, 17B	18A, 18B	CH 15, 2W/4WR CH 15, 4WT	CH 17, 2W/4WT CH 18, 2W/4WT
1	19A, 19B	20A, 20B	CH 16, 2W/4WR CH 16, 4WT	CH 19, 2W/4WT CH 20, 2W/4WT
1	21A, 21B	22A, 22B	*CH 17, 2W/4WR *CH 17, 4WT	CH 13, 2W/4WT CH 14, 2W/4WT
1	23A, 23B	24A, 24B	*CH 18, 2W/4WR *CH 18 4WT	CH 15, 2W/4WT CH 16, 2W/4WT
2	1A, 1B	2A, 2B	*CH 19, 2W/4WR *CH 19, 4WT	CH 17, 2W/4WT CH 18, 2W/4WT
2	3A, 3B	4A, 4B	*CH 20, 2W/4WR *CH 20, 4WT	CH 19, 2W/4WT CH 20, 2W/4WT

*COMPLETE AFTER TEST OF CH 16

b. Repeat step (a) with the roles of the telephones interchanged.

c. Repeat steps a and b for SIG 3 and SIG 5.

d. Repeat steps a and b for SIG 1 and SIG 5.

e. Repeat steps a and b for SIG 7 and SIG 9.

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4.7.2.4 4 Wire Trunks.-

a. Connect a J-1077 box to the Sig 2 U-187 connector at the signal entrance panel.

b. Connect the J-1077 binding posts corresponding to CH 21 transmit to CH 22 receive and from CH 21 receive to CH 22 transmit (SEE Table VIII).

c. Repeat step b from Channel 25 and 26, Channel 27 and 28, and Channel 29 and 30 (or any other 4 wire trunk pairs).

d. Connect TA-938 Telephones to any two of the DTMF Channels (Channels with Type II Terminal Cards).

e. Key 123 on one of the TA-938 phones, a second dial tone will be heard in the receiver. Key IXX (the number of the second TA-938). The other phone shall ring. Pick up the handset of the called telephone, a voice patch shall be completed and communication between the two phones shall be possible.

f. Repeat step e. for channel 23 and 24, channel 25 and 26, channel 27 and 28 and channel 29 and 30.

g. Repeat steps a thru f for all other even numbered signals connected to a switchboard.

TABLE VIII

Channel #	Binding Post	Channel #	Binding Post
21 receive	5a, 5b	22 transmit	8a, 8b
21 transmit	6a, 6b	22 receive	7a, 7b
23 receive	9a, 9b	24 transmit	12a, 12b
23 transmit	10a, 10b	24 receive	11a, 11b
25 receive	13a, 13b	26 transmit	16a, 16b
25 transmit	14a, 14b	26 receive	15a, 15b
27 receive	17a, 17b	28 transmit	20a, 20b
27 transmit	18a, 18b	28 receive	19a, 19b
29 receive	21a, 21b	30 transmit	24a, 24b
29 transmit	22a, 22b	30 receive	23a, 23b

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4.7.3 Spare Patch Panel Tests.-

a. Connect TA-938 telephones respectively to pins 1A, 1B, and 2A, 2B of SPARE SIGNAL 1 junction box. Patch from corresponding jacks at line side of patch panel to CH 11 and CH 12 jacks of SIG 1 at switchboard side of patch panel. Use one telephone to call the other by keying (dialing) IXX. The called telephone shall ring, pick up telephone. A voice patch shall be completed, and communication between the telephones shall be possible.

b. Repeat step a for all other spare jacks on the line side of the patch panel. Test two spare jacks at a time.

4.7.4 TA-312 Telephone Conference Jacks Test.-

a. Connect TA-312 telephones to pins 25A and 25B of SIG 1 thru SIG 5 junction boxes. Patch end of these SIGNAL phone jacks and the TA-312 phone jack randomly to the group 1 conference jacks. Communication among all six telephones shall be possible.

b. Repeat step a for SIG 2 thru SIG 6 (SIG 6 thru SIG 10 in 120 line assemblage) and group 2 conference jacks.

4.7.5 Headset Switch Tests.- (Applicable to 120-line assemblage only)

a. Disconnect headsets from MASTER switchboards. Connect headset cables from the switchboards to the switchbox. Reconnect one of the headsets to the switchbox.

b. Connect TA-312 telephones to Ch 1 pins of SIG 1 and SIG 7 junction boxes, i.e. to pins 1A and 1B of each junction box.

c. Crank the SIG 1 telephone. The CALL ANS button on the MASTER switchboard of Right shall light. Place the headset switch in the SYSTEM 1 position, and depress the CALL ANS button. A voice path shall be completed, and communications between the SIG 1 telephone and the headset shall be possible.

d. Place the headset switch in Left position. The voice patch previously established shall be terminated.

e. Repeat steps c and d, with the roles of SIG 1 and SIG 7 telephones interchanged.

4.8 Vehicular Transportation test.- The shelter facility shall be secured (and blocked to prevent shifting) to a government furnished 1 1/4 ton, M-715, truck and transported over the specially prepared course described in para 4.9.1 for 10 laps at each of the speeds 5 mph, 10, mph, 15 mph and 18-20 mph.

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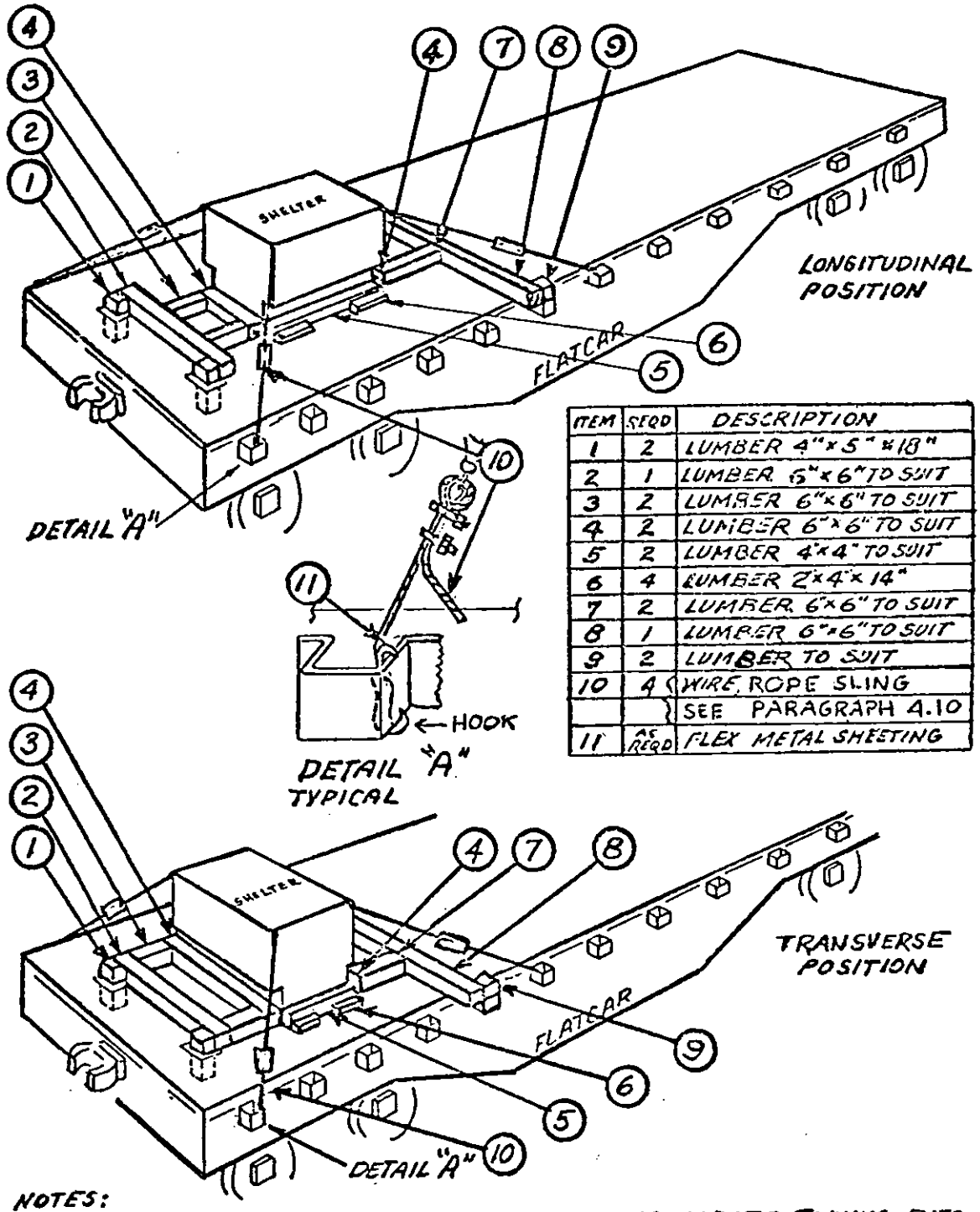
A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.5 after completion of the tests (Note: The shelter facility shall be loaded to simulate the AN/TTC-41(V) using live loads). (See 6.1).

4.8.1 Vehicular Transportation Course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have 1 x 1 inch chamfers. The sixth and twelfth timber shall be placed at 45 degrees to the travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored down securely by any effective means. The surface on which the timbers are placed shall be hard and level.

4.9 Rail Transport, Impact Test.- The shelter facility loaded as specified in paragraph 4.9 secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tie down slings, shall be as specified in Figure 1 and shall meet the provisions of the test as specified in para 4.10.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. The test shall consist of four impacts. Two impacts with the shelter facility positioned logitudinally on the flatcar, one from each end of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.9.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in figure 1. All other blocking shall be firmly secured as shown in the figure.

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ITEM	REQD	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 6"x6" TO SUIT
5	2	LUMBER 4"x4" TO SUIT
6	4	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE ROPE SLING SEE PARAGRAPH 4.10
11	AS REQD	FLEX METAL SHEETING

NOTES:
1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.
2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

Figure 1. Railcar Tiedown and Blocking

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Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.9.2 Loaded Gondola Car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.10 Drop tests.-

4.10.1 Flat drop.- The shelter facility loaded as in para 4.9 will be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of para 3.7.

4.10.2 Rotational drop.- The shelter facility loaded as in para 4.9 shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of para 3.7.

4.11 Watertightness Test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (NOTE: Perfrom tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G 29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer.

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(See Figure 2 for nozzle location.) Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of para 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel.) Upon completion of the above test, the shelter facility shall meet the requirements of para 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the exhaust blower vent cover, the battery box vent cover and the door vent panel at the rear of the shelter) and repeat a above.

4.12 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 26 inches (measured from the bottom of the skids) and meet the requirements of para 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tie downs. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

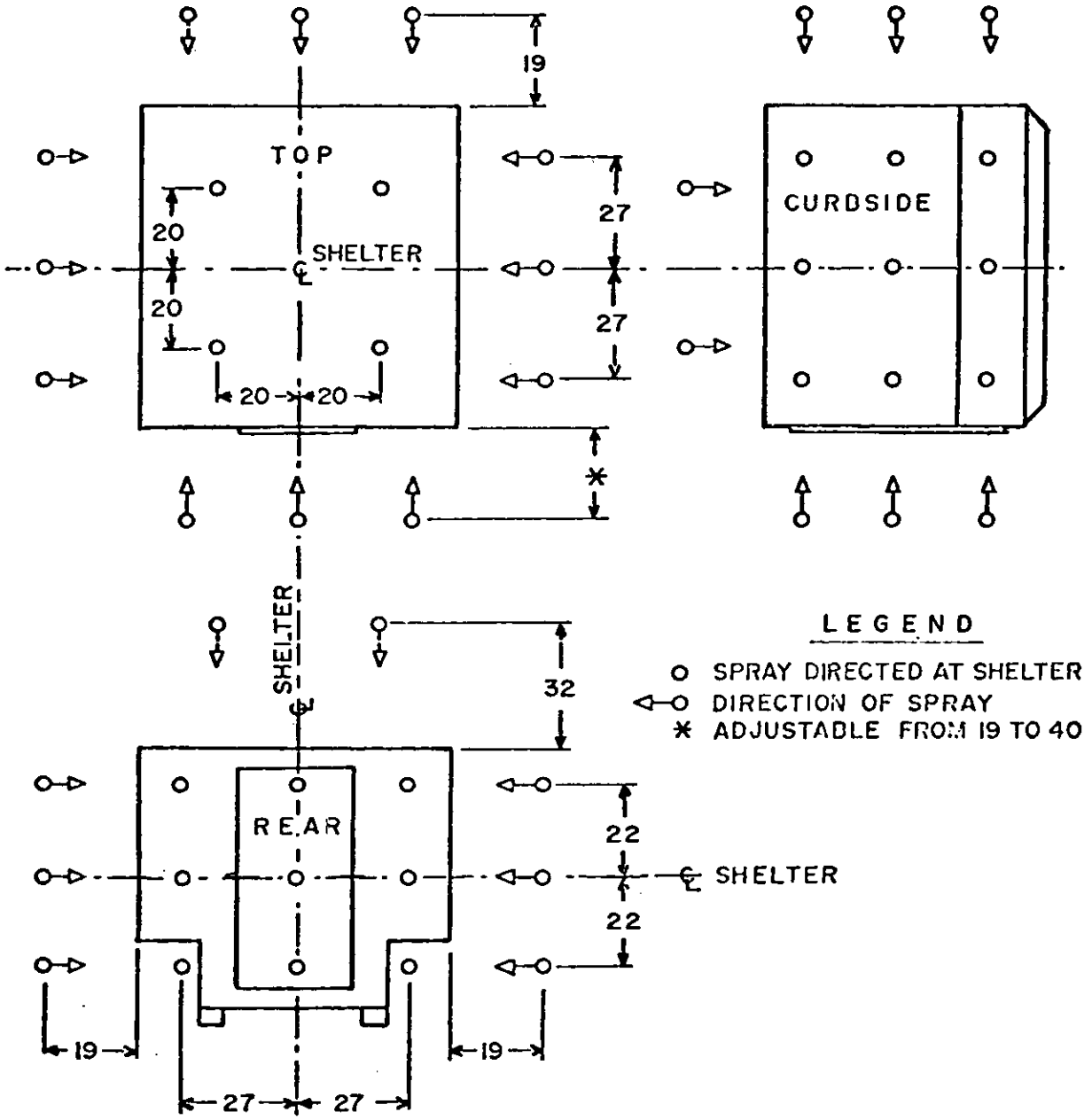
4.13 Humidity test.- A hardware sample of five of each size and type (if the total quantity of any size or type is less than five, all shall be tested) of bolt, screw, nut, flat washers, lock washers and rivnuts shall be randomly selected for test. The hardware sample shall be subject to the humidity test specified in Method 507, Procedure II of MIL-STD-810 to determine conformance to the requirements of paragraph 3.10.

4.14 Mounting insert tests.-

4.14.1 First Article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table VII. After test the requirements of paragraph 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Group B insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected in each sample equipment. Each selected insert shall be tested in accordance with Table VII. After test the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

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NOTES

1. NOZZLES SHALL BE PERPENDICULAR TO SHELTER SURFACE.
2. UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3. GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4. ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5. FRONT NOZZLE LOCATIONS SAME AS REAR.
6. DIMENSIONS ARE IN INCHES.

Figure 2. Nozzle Locations

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TABLE IX Mounting Insert Loads

Insert	First Article and Group C Test		Group B Test	
	Axial (Lbs)	Torque (inch Lbs)	Axial (Lbs)	Torque (inch Lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.15 Acoustical noise test.- The shelter facility shall be located in an area remote from excessive machinery, vehicle and other noise operating sources to reduce ambient noise to a minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band Analyzer, General Radio Corporation, Model 1558A, or equal, and Microphone, Piezoelectric, General Radio Corporations, Model 1560-P6 or equal, record ambient noise levels over the frequency ranges specified in 3.12 at the positions specified in paragraph 4.16.1. Operate the air conditioner in the cooling, high fan speed mode, and repeat noise level recordings at the same frequency levels and positions as above. Repeat above procedure with the exhaust blower operating and the air conditioner off. The noise levels, corrected from ambient shall meet the requirements of 3.12.

4.15.1 Measurement location.- Noise measurements shall be made at the following location:

Operators position, 28" from front wall, 26" from curbside wall, 32" above floor.

4.16 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lug shall be tested for conformance to 3.22.4.

4.17 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.24. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.18 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.25 and Table VIII.

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TABLE X - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. Inspection of welded assemblies, such as equipment racks, entrance boxes, etc., should be made prior to final paint application. 4. Sealer improperly applied (holes, separations or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc., inoperative. 9. Electrical receptacles, switches, jacks, connectors, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16". 12. The blower housing does not align properly with the mounting inserts of the shelter facility. 13. The air conditioner cover does not align properly with the mounting inserts of the shelter facility. 14. Ground stud not properly sealed. 15. Bulkhead Power and Signal connector fastners not sufficiently tight. 16. Electrical shematics missing.
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges on bracket not removed 5. Screws loose. 6. Gaskets, not completely and securely adhered. 7. Electrical schematics not properly adhered to wall.

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4.19 Packaging inspection.- Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.19.1 First Article Inspection Packaging.- Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.19.2 Quality Conformance Inspection of Packaging.-

4.19.2.1 Materials Inspection.- All materials to be used in packaging shall be inspected in accordance with the applicable material specifications.

4.19.2.2 Preservation Inspection.- Inspection of preservations and interior markings shall be in accordance with the group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.19.2.3 Packaging Inspection.- Inspection of packing and the marking for shipment and storage shall consist of the examination specified in Table XI, "PACKING INSPECTION PROVISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

TABLE XI PACKING INSPECTION PROVISIONS

NO.	CHARACTERISTICS	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specifictions, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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4.20 Functional tests after rough handling tests.- After completion of the rough handling tests specified in 4.19, the power and signal wiring test (See 4.6) shall be conducted to determine freedom from operational malfunction caused by rough handling.

4.21 SEL Form 1183.- An inspection shall be performed, on one unit only, during First Article testing, SEL Form 1183, to verify compliance with Para 3.21 (System Safety Engineering).

5. PACKAGING

5.1 Packaging requirements.- The requirements for packaging shall be in accordance with MIL-S-55507.

6. NOTES

6.1 Intended Use.- The Shelter Electrical Equipment S-561()/TTC-41(V) is the shelter facility portion of the AN/TCC-41(V)1, (V)2, (V)3, (V)4, and it is used to house the following operational equipments:

	(V)1	(V)2	(V)3	(V)4	Equipments
a.	1 ea.	2 ea.	3 ea.	4 ea.	SB-3614()/TT Switchboard
b.	1 ea.	1 ea.	1 ea.	2 ea.	PP-6224/U Power Supply
c.	1 ea.	1 ea.	1 ea.	1 ea.	LS-147() FI Intercom
d.	1 ea.	1 ea.	1 ea.	1 ea.	TA-312/PT Telephone
e.	1 ea.	1 ea.	1 ea.	1 ea.	TA-938/ G Telephone

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads or the government furnished live loads (operational equipment) as specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-561()/TTC-41(V).

d. A list of equipments to government loaned for testing purposes.

e. Level A or B preservation and packing (See Section 5).

f. When first article inspection rough handling tests are not required.

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g. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special test equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.4)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-561R/TTC-41(V). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Tropicalization of material.- All materials of construction are specified on the drawings and are Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus-Resistance satisfactorily in accordance with requirement 508 of MIL-STD-810.

6.8 Level B preservation.- When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.9 Environmental.- Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.10 Subject term (Key word) listing.

AN/TTC-41
Central office, telephone automatic
S-561( )/TTC-41(V)
Shelter, electrical
Shelter facility

6.11 Changes from previous issue.- The margins of this standard are marked with vertical lines to indicate where significant changes from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations.

Custodian:
Army - CR

Preparing Activity:
Army - CR
(Project 5805-A315)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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